QSAR-Guided Generative Design for TB Drug Discovery

Model Evaluation, Series C/E Results, and Generative Campaign Outputs

Destinee Zhu Roy

TB QSAR Project

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Outline

1. QSAR Model Development
2. Series C Evaluation
3. Series E Prospective Predictions
4. ADME and Physicochemical Properties
5. Generative Modes
6. Gate 1 Shortlist
7. Generation Output Animations
8. Summary

What We Built: A QSAR Predictive Model

Goal: Predict IC_{50} (potency vs. Mtb) from molecular structure alone

Input – Molecular Descriptors

- **RDKit** physicochemical descriptors (primary)
- **ECFP_r3** extended-connectivity fingerprints
- **MACCS** structural keys
- **Mordred** descriptor library

Output

$\hat{y} = \log_{10}(IC_{50}, \mu M)$
converted to pIC_{50} and IC_{50} for display

Workflow

1. Load Series C/E experimental data
2. Compute descriptor matrices
3. Train / cross-validate multiple regressors
4. Select best model (highest CV R^2)
5. Build ensemble for uncertainty
6. Save artifact `qsar_best_model.joblib`

Model Selection: Candidates and Results

Descriptor	Regressor	CV R^2	Role
RDKit	K-Nearest Neighbours (KNN)	**0.887**	Primary model
ECFP_r3	K-Nearest Neighbours (KNN)	0.677	Ensemble #1
RDKit	Decision Tree (DT)	0.665	Ensemble #2
RDKit	LightGBM (LGBM)	0.540	Ensemble #3
RDKit	Extra Trees (ET)	0.433	Ensemble #4

Why RDKit + KNN?

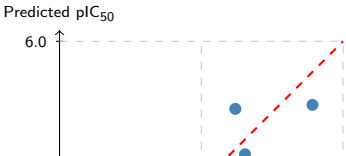
- Highest cross-validated R^2 on training data
- Robust to small dataset size (low-data QSAR regime)
- Interpretable: prediction is distance-weighted average of nearest neighbours

Ensemble Purpose

- Four additional models predict in parallel
- Uncertainty $= \sigma$ of ensemble predictions
- Flags structurally novel molecules where primary model may extrapolate

Series C: Model Evaluation on Test Compounds

Compound ID	Measured pIC_{50}	Predicted pIC_{50}	Residual
ROY-0000173-001	5.328	5.302	−0.026
ROY-0000174-001	5.009	5.183	+0.174
ROY-0000220-001	6.107	5.931	−0.176
ROY-0000221-001	— (¿200 μM)	4.848	—
ROY-0000175-001	5.165	5.215	+0.050
ROY-0000211-001	5.745	5.909	+0.164
5 measurable compounds			



Performance Metrics

- **MAE** = 0.118 pIC_{50} units
- $\mathbf{R}^2$ = 0.887

Series C: IC_{50} Comparison

Compound ID	Measured IC_{50} (μM)	Predicted IC_{50} (μM)	Series
ROY-0000173-001	4.697	4.98	C Spiro
ROY-0000174-001	9.786	6.57	C Spiro
ROY-0000220-001	0.781	1.17	C Spiro
ROY-0000221-001	>200	14.18	C Spiro
ROY-0000175-001	6.843	6.09	C Spiro
ROY-0000211-001	1.800	1.23	C Spiro

Key Observations

- Model correctly identifies ROY-0000220-001 and ROY-0000211-001 as most potent (sub-2 μM predicted)
- ROY-0000221-001 has measured IC_{50} >200 μM but predicted at 14.18 μM; structurally distinct (indole side-chain) — model underestimates inactivity for novel scaffolds
- Rank order of active compounds is well preserved

Series E: Prospective Predictions (No Measured Data)

Compound ID	Pred. pIC_{50}	Pred. IC_{50} (μM)	Cluster	Note
ROY-0000308	4.883	13.08	1	Fluorophenyl
ROY-0000307	4.467	34.11	1	Bromopyridine (weakest)
ROY-0000287	5.132	7.38	−1	Hexyloxy (best predicted)
ROY-0000286	4.953	11.15	−1	Phenylpropyloxy
ROY-0000275	4.563	27.33	1	Bromopyridine
ROY-0000268	4.767	17.09	1	Alkyne-phenyl
Range		7.38 – 34.11 μM	pIC_{50}: 4.47 – 5.13	

Structural Features

- All contain spiro-piperidine (Boc-protected) core
- ROY-0000287 (hexyloxy chain) has best predicted potency
- Halopyridine compounds predicted as

Prediction Uncertainty

- No measured IC_{50} available for Series E
- Ensemble uncertainty = 0 (uniform predictions — extrapolation region)
- Experimental validation needed

ADME Prediction Properties

Properties Predicted per Molecule

Property	Method
cLogP	RDKit Crippen
logS (solubility)	ESOL model
Microsomal CL	AZ QSAR model
Hepatocyte CL	AZ QSAR model
Half-life	Obach model
Human oral absorption	QikProp
Caco-2 permeability	QikProp

Gate 1 Shortlist ADME Summary

Property	Range
pred_cLogP	1.93 – 2.80
pred_logS_ESOL	−5.34 to −3.51
ADME indiv. score	0.70 – 0.94
passes_adme_indiv	65/200 (33%)
passes_metab. stab	65/200 (33%)

65 of 200 shortlisted compounds pass all ADME+metabolic stability filters — these are highest-priority compounds for synthesis

Generative Modes: Three Strategies

Mode	Description	Output Characteristic
Plain (Sampling)	Sample directly from the prior model without any optimization signal	Maximally diverse; explores full chemical space of the prior
Reinforcement Learning (RL)	Score each generated molecule; reward high-scoring structures to shift the distribution	Focused; converges on potency/ADME optima; risk of mode collapse
Curriculum Learning (CL)	Progressive reward shaping — start broad, tighten criteria over epochs	Balanced diversity and optimisation; avoids mode collapse

Molecular Transformation Modes

- **mol2mol_similarity** — generate analogues with Tanimoto constraint
- **mol2mol_scaffold** — preserve Murcko

Generator Families Used

reinvent	SMILES-based RNN
reinvent_pubchem	PubChem-pretrained
libinvent	R-group decoration

Generation Campaign: Mode × Family Matrix

Family	Plain	RL	CL	mol2mol
reinvent	✓	✓	✓	—
reinvent_pubchem	✓	✓	✓	similarity
libinvent	—	—	✓	—
libinvent_transformer_pubchem	—	—	✓	—
linkinvent	—	—	✓	—
linkinvent_transformer_pubchem	—	—	✓	similarity
pepinvent	✓	✓	—	—
nitro_bioisostere	✓	✓	✓	scaffold_generic

Master Pool Statistics

Total candidates	37,451
Unique scaffolds	14,287
Source families	8
Scoring models	QSAR + ADME

Scoring per Candidate

- pred_pIC50 (primary QSAR)
- pred_cLogP, pred_logS_ESOL
- Microsomal / hepatocyte clearance
- Tanimoto similarity to training set

Gate 1 Shortlist: 200 Top Candidates

Triage Score Formula

$$\text{triage} = w_1 \cdot \hat{p}\text{IC}_{50} + w_2 \cdot \text{ADME} + w_3 \cdot (1 - T_{\max})$$

Balances potency, drug-likeness, and novelty

Property	Min	Max
Triage score	0.770	0.921
pred pIC_{50}	4.94	6.11
pred IC_{50} (μM)	0.78	11.55
Tanimoto (to train)	0.28	0.86
pred cLogP	1.93	2.80
pred logS ESOL	−5.34	−3.51

Source Family (200 compounds)

Family	Count
reinvent_pubchem	137
nitro_bioisostere	53
reinvent	9
linkinvent_trans.	1

Source Mode (200 compounds)

Mode	Count
plain	140
mol2mol_similarity	28
mol2mol_scaffold_generic	22
curriculum_learning	6
mol2mol_scaffold	3

Gate 1 Shortlist: Unique Scaffolds

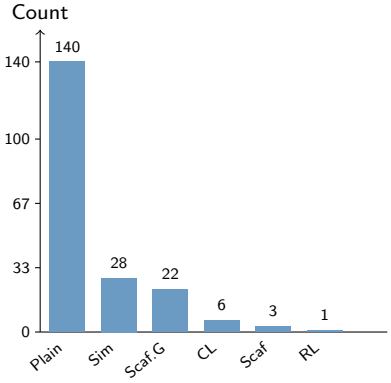
Scaffold Diversity

	Master Pool	Gate 1
Candidates	37,451	200
Unique scaffolds	14,287	155
Scaffold diversity	38.1%	77.5%

Gate 1 shortlist has **much higher scaffold diversity** than the master pool — the triage score effectively up-weights novel chemotypes.

Next Step

Compounds passing `passes_adme_individual` (65 of 200) are highest-priority for experimental synthesis and testing.

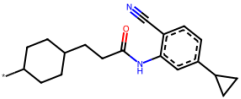
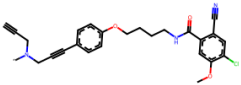
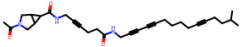
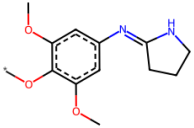
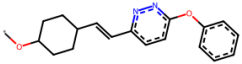
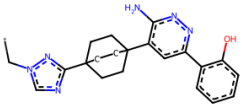


Source mode distribution of Gate 1 shortlisted compounds

Campaign Overview: All Generation Folders

All Generation Folders: Plain vs RL vs CL (time-evolving)

Step 1/10

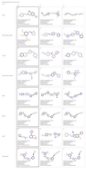
	Plain	RL	CL
libinvent	 Pred IC50: 13.86 uM | Actual IC50: n/a uM Pred pIC50: 4.86 | MW: 295.41 cLogP: 4.68 | logS: -5.26 Mic: 94.85 | Hep: 100.11 Half-life: 45.63 | Sol: Moderately Soluble	 Pred IC50: 15.34 uM | Actual IC50: n/a uM Pred pIC50: 4.81 | MW: 448.93 cLogP: 3.56 | logS: -5.47 Mic: 108.50 | Hep: 96.27 Half-life: 93.74 | Sol: Moderately Soluble	 Pred IC50: 8.29 uM | Actual IC50: n/a uM Pred pIC50: 5.08 | MW: 489.66 cLogP: 2.73 | logS: -4.97 Mic: 34.61 | Hep: 28.09 Half-life: 50.31 | Sol: Moderately Soluble
libinvent_transformer_pubchem	 Pred IC50: 21.40 uM | Actual IC50: n/a uM Pred pIC50: 4.67 | MW: 235.26 cLogP: 1.96 | logS: -2.88 Mic: 18.65 | Hep: 18.04 Half-life: 1.22 | Sol: Soluble	 Pred IC50: 11.42 uM | Actual IC50: n/a uM Pred pIC50: 4.94 | MW: 295.36 cLogP: 4.32 | logS: -5.10 Mic: 113.15 | Hep: 120.47 Half-life: 36.77 | Sol: Moderately Soluble	 Pred IC50: 18.66 uM | Actual IC50: n/a uM Pred pIC50: 4.73 | MW: 375.46 cLogP: 3.07 | logS: -4.74 Mic: 47.32 | Hep: 16.28 Half-life: 4.30 | Sol: Moderately Soluble

Top Generation Outputs: Combined View

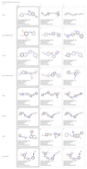
Combined Animation Board: all requested outputs

Frame 1/13

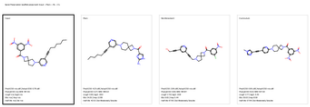
all_generation_folders_big_timeline_first_frame.png



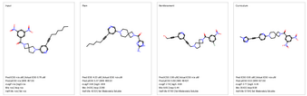
all_generation_folders_big_timeline.gif



spiral_preservation_big.gif



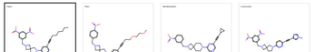
spiral_preservation_all_compounds_part_01.gif



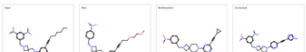
scaffold_preservation_minimal.gif

scaffold_non_preservation_minimal.gif

non_spiral_preservation_big.gif



non_spiral_preservation_all_compounds_part_01.gif



minimal_combined_summary.gif

End-to-End Generation Pipeline: Phase Overview

Everything In One GIF (Phased)

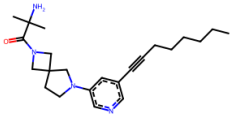
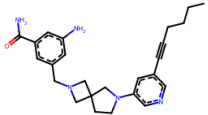
Phase 1: Global Generation Overview

All folders, synchronized Plain / RL / CL timeline

Selected Folders: Detailed Output

Selected Folders Only (within requested paths)

Step 1/14

	Plain	RL	CL
nitro/mol2mol_mmp	Plain  Pred IC50: 4.72 uM | Actual IC50: n/a uM Pred pIC50: 5.33 | MW: 382.55 cLogP: 3.18 | logS: -4.70 Mic: 44.50 | Hep: 36.57 Half-life: 42.22 | Sol: Moderately Soluble	RL No data	CL No data
nitro/mol2mol_scaffold	Plain  Pred IC50: 3.53 uM | Actual IC50: n/a uM Pred pIC50: 5.45 | MW: 417.56 cLogP: 3.02 | logS: -4.88 Mic: 23.22 | Hep: 26.07 Half-life: 42.78 | Sol: Moderately Soluble	RL No data	CL No data
	Plain	RL	CL

Summary: QSAR Model Performance

Metric	Value	Context
Primary model	RDKit + KNN	Highest CV R^2
CV R^2	0.887	Excellent for low-data QSAR
Test MAE (Series C)	0.118 pIC_{50}	≈32% IC_{50} error
Test R^2 (Series C)	0.887	5 measured compounds
Series E predictions	7.38 – 34.11 μM	6 prospective compounds
Ensemble models	4 additional	Uncertainty estimation

Generation Campaign

Total candidates	37,451
Gate 1 shortlisted	200
Unique scaffolds (Gate 1)	155
ADME-passing	65 (33%)
Best pred IC_{50}	0.78 μM
Best triage score	0.921

Recommended Next Steps

1. Synthesise top 65 ADME-passing Gate 1 compounds
2. Measure IC_{50} vs. Mtb H37Rv
3. Re-train QSAR with new data (Series C + E + Gate 1 actives)
4. Iterate: run new generative campaign

All Prediction Measurements at a Glance

Measurement	Gate 1 Range	Description
pred_pIC50	4.94 – 6.11	Primary potency prediction
pred_ic50_uM	0.78 – 11.55 μM	IC_{50} in micromolar
pred_cLogP	1.93 – 2.80	Lipophilicity (Crippen)
pred_logS_ESOL	-5.34 to -3.51	Aqueous solubility (log mol/L)
pred_MetStab_CL_Microsome	—	Microsomal clearance (AZ model)
pred_MetStab_CL_Hepatocyte	—	Hepatocyte clearance (AZ model)
adme_individual_score	0.70 – 0.94	Composite ADME scoring
triage_score	0.77 – 0.92	Final composite ranking
Tanimoto	0.28 – 0.86	Similarity to training set
pred_pIC50_uncertainty	0 (current run)	Ensemble std (σ predictions)

All predictions

generated by the primary RDKit+KNN QSAR model and AZ ADME models. IC_{50} uncertainty requires a re-run with full ensemble enabled.

Thank You

QSAR-Guided Generative Design for TB Drug Discovery

March 2026

Primary model:	RDKit + KNN, $R^2 = 0.887$
Test MAE:	0.118 pIC_{50} (Series C)
Shortlisted:	200 compounds from 37,451
Best predicted:	0.78 μM IC_{50}